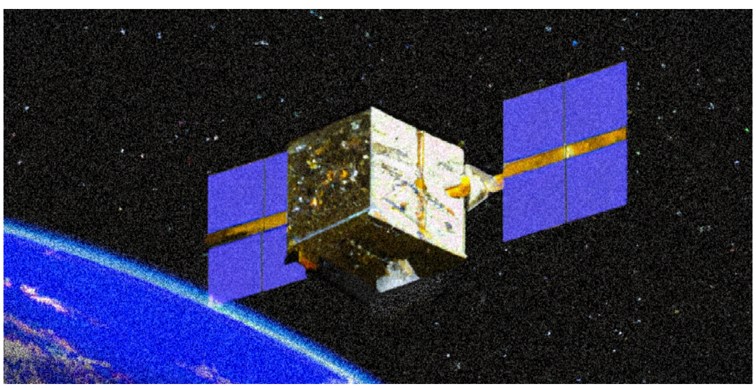
TTK4190 Assignment 1
Kinematics and Kinetics

Objective
You will learn how to implement and simulate the attitude kinematics and kinetics of a rigid body in space. The theory is also applicable to underwater vehicles, robotic systems, and uncrewed aerial vehicles.

Delivery Details
You must deliver a report and produce your own Matlab files for each simulation problem. Simulink should not be used. Matlab m-files should not be included in the delivery. Ensure all plots clearly show the required data (title, legend, label, tag the axes, set the grid on, and so forth). An example of how to simulate the attitude dynamics of a rigid body (using unit quaternions) is demonstrated in the file attitude.m, which is posted on Blackboard. Using this file as a template and changing the necessary parts of the code is recommended. To run the template, you have to install the MSS toolbox (Marine Systems Simulator). The easiest way to do this is to install it as an Add-On, link here.

The report has to be handed in via Blackboard. You should write the report on your PC using your favorite editor (LaTeX, Word, Pages...). It is also possible to hand in a scanned version, but in the end, it has to be in PDF format. **Paper versions are not accepted**.



Consider a cube-shaped satellite with sides of length $L = 0.1\,\text{m}$. The inertia tensor for such a satellite with axes of rotation through its center is

$$\mathbf{I}_g^{CG} = \mathbf{I}_g^b = m \begin{bmatrix} R_{44}^2 & 0 & 0 \\ 0 & R_{55}^2 & 0 \\ 0 & 0 & R_{66}^2 \end{bmatrix} \tag{1}$$

where $m = 10$ kg is the mass of the satellite and $R_{44} = R_{55} = R_{66} = L/\sqrt{6}$ m are the radii of gyration with respect to the CG. The attitude dynamics of this satellite can be described by the following quaternion-based equations of motion:

$$\begin{aligned} \dot{\mathbf{q}} &= \mathbf{T}_q(\mathbf{q})\omega \\ \mathbf{I}_g\dot{\omega} - \mathbf{S}(\mathbf{I}_g\omega)\omega &= \tau \end{aligned} \tag{2}$$

Problem 1: Attitude Control of Satellite

1.1 What is the equilibrium point $\mathbf{x}_0$ of the open-loop system $\mathbf{x} = [\varepsilon^\top, \omega^\top]^\top$ corresponding to $\mathbf{q} = [\eta, \varepsilon_1, \varepsilon_2, \varepsilon_3]^\top = [1, 0, 0, 0]^\top$ and $\boldsymbol{\tau} = \mathbf{0}$? (To clarify the construction of the state vector; it is not necessary to include the state η since it is implicitly included as function of ε, i.e., $\pm\eta = \sqrt{1 - \varepsilon^\top \varepsilon}$.)

Linearize the spacecraft model about $\mathbf{x} = \mathbf{x}_0$ and write down the expressions for the $\mathbf{A}$ and $\mathbf{B}$ matrices of the linearized dynamics,

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{\epsilon} \\ \dot{\omega} \end{bmatrix} = \mathbf{A}\mathbf{x} + \mathbf{B}\tau \tag{3}$$

Hint: use eq. (2.77) in (Fossen, 2021)

1.2 Consider the attitude control law:

$$\tau = -\mathbf{K}_d \omega - k_p \epsilon \tag{14}$$

where $k_p > 0$ is a scalar control gain, $\mathbf{K}_d = k_d\, \mathbf{I}_3$ is a controller gain matrix with $k_d > 0$, and ϵ is the imaginary part of the unit quaternion. Let $k_p = 5 \times 10^{-5}$ and $k_d = 9 \times 10^{-4}$ and verify that the linearized closed-loop system is stable.

Would you prefer real or complex poles in this particular application? Explain why/why not.

1.3 Let $k_p = 5 \times 10^{-5}$ and $k_d = 9 \times 10^{-4}$. Modify attitude.m and simulate the attitude dynamics in eq. (2) of the closed-loop system with the control law given by eq. (14) with initial conditions set to:

$$[\phi(0) = -5^o\,,\, \theta(0) = 10^o\,,\, \psi(0) = -20^o]$$

The initial angular velocities should be zero. Plot the resulting attitude in Euler angles.

Does the behavior of the system match your expectations? Explain why/why not. How would you modify the control law to follow nonzero constant reference signals?

Include figures of the Euler angles, angular velocities, and the control inputs in the report.

1.4 Consider the modified attitude control law:

$$\tau = -\mathbf{K}_d \omega - k_p \tilde{\epsilon} \tag{17}$$

where $\tilde{\epsilon}$ is the error in the imaginary part of the quaternion (between the desired and true state) defined as:

$$\tilde{\mathbf{q}} := \begin{bmatrix} \tilde{\eta} \\ \tilde{\epsilon} \end{bmatrix} = \bar{\mathbf{q}}_d \otimes \mathbf{q} \tag{18}$$

where $\mathbf{q}_d$ is the desired quaternion, $\mathbf{q}$ is the current state, and $\bar{\mathbf{q}} = [\eta, -\epsilon^\top]^\top$ denotes the conjugate (sometimes called the inverse) of a quaternion $\mathbf{q}$. Furthermore, the quaternion product is given as:

$$\mathbf{q}_1 \otimes \mathbf{q}_2 = \begin{bmatrix} \eta_1\eta_2 - \epsilon_1^\top \epsilon_2 \\ \eta_1\epsilon_2 + \eta_2\epsilon_1 + \mathbf{S}(\epsilon_1)\epsilon_2 \end{bmatrix} \tag{19}$$

where $\mathbf{S}(\epsilon_1)$ is the skew-symmetric matrix.

Write down the matrix expression for the quaternion error $\tilde{\mathbf{q}}$ on the component form. What is $\tilde{\mathbf{q}}$ after convergence, is $\mathbf{q} = \mathbf{q}_d$?

1.5 Now change the gains to $k_p = 5 \times 10^{-4}$ and $k_d = 9 \times 10^{-3}$. Simulate the attitude dynamics of the closed-loop system with the control law given by eq. (17). The desired attitude is given by the time-varying reference signal $\mathbf{q}_d(t)$ corresponding to Euler angles given in degrees as:

$$[\phi(t) = 0\,,\, \theta(t) = 15\cos(0.1t)\,,\, \psi(t) = 10\sin(0.05t)]$$

Set the initial conditions equal to those in Problem 1.3. Does the behavior of the system match your expectations? Explain why/why not.

Include the same set of figures as in Problem 1.3 and add a graph of the tracking error in the report.

1.6 Consider the modified attitude control law:

$$\tau = -\mathbf{K}_d\tilde{\omega} - k_p\tilde{\epsilon} \tag{20}$$

where $\tilde{\boldsymbol{\omega}} = \boldsymbol{\omega} - \boldsymbol{\omega}_d$ is the difference between the desired and current angular velocity. Let the reference signals from Problem 1.5 give the desired attitude and calculate the desired angular velocity using,

$$\omega_d = \mathbf{T}_{\Theta}^{-1}(\Theta_d)\dot{\Theta}_d$$

where Θ_d is the desired Euler angle. See eq. (2.39) in Fossen (2021) for more details. Simulate the control law in eq. (20) with the same gains, initial conditions, and reference signals as in Problem 1.5. Deliver the same set of figures and compare the results.

Does the behavior of the system match your expectations? Explain why/why not. Can you propose a way to improve the control law further?

1.7 In this problem, you will use Lyapunov analysis to analyze the stability of a closed-loop system. Assume that the control law given by eq. (17) is used to perform setpoint regulation, such that in the equilibrium, the states of the system are: $\omega_d = \mathbf{0}$, $\epsilon_d =$ constant, $\eta_d =$ constant.

Use the following Lyapunov function candidate to perform the analysis (Fjellstad and Fossen, 1994):

$$V = \frac{1}{2}\tilde{\omega}^{\top}\mathbf{I}_g\tilde{\omega} + 2k_p(1 - \tilde{\eta}) \tag{25}$$

Moreover, use this additional information in your analysis. It can be shown that:

$$\dot{\tilde{\eta}} = -\frac{1}{2}\tilde{\epsilon}^{\top}\tilde{\omega} \tag{26}$$

a) Show that the time derivative of the Lyapunov function is:

$$\dot{V} = -k_d\omega^{\top}\omega \tag{27}$$

b) Use a suitable Lyapunov theorem/method to prove that the equilibrium of the closed-loop system is asymptotically stable.

Hint: check theorem 4.1 in (Khalil, 2002)

c) Can you prove global asymptotic stability? Explain.

References

Fossen, T.I. (2021) Handbook of Marine Craft Hydrodynamics and Motion Control, 2nd Edition. John Wiley & Sons.

Fjellstad, O.-E. and Fossen, T.I. (1994). Quaternion Feedback Regulation of Underwater Vehicles. 3rd IEEE Conference on Control Applications (CCA), Glasgow, UK, IEEE Xplore, pp. 857-862. doi.org/10.1109/CCA.1994.381209

Khalil, H.K. (2002) Nonlinear Systems, 3rd Edition. Prentice-Hall.

C. G. Mayhew, R. G. Sanfelice, and A. R. Teel. On quaternion-based attitude control and the unwinding phenomenon, Proceedings of the 2011 American Control Conference. IEEE, 2011. p. 299-304.

Mahony, R., Hamel, T., Pflimlin, J. M. (2008). Nonlinear complementary filters on the special orthogonal group. IEEE Transactions on automatic control, 53(5), 1203-1218. doi.org/10.1109/TAC.2008.923738